Real-Time Subsurface Utility Detection in GPR Data Using Region Vectorization and Inverted Parabolic Approximation

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***Abstract*—Locating buried utilities with Ground Penetrating Radar (GPR) typically requires an operator to interpret hyperbolic reflections in B-scan images. We present a signal-feature-based method that detects subsurface utilities directly from incoming A-scans, without a labelled training set or a hyperbola shape search. Each A-scan is segmented into regions defined by consecutive extrema, with each region described by its polarity pattern, extrema spacing, and amplitude. Regions with similar features are tracked across neighbouring traces to form coherent reflection trajectories, which are approximated with parabolic models to localize buried utilities. The method runs as a streaming pipeline and is evaluated on 194 labelled surveys acquired with a Noggin 250 MHz GPR system, compared against C3 clustering and a Hough transform detector under identical conditions. On 101 held-out surveys containing 197 labelled utilities, the proposed method reaches an F1 score of 0.57, comparable to C3 (0.61) and the Hough transform (0.55), with the highest recall among the three methods (0.62) and the lowest processing cost (0.07 ms per A-scan, 5–6× faster than the baselines).**

***Keywords*—Ground Penetrating Radar (GPR), subsurface detection, region vectorization, inverted parabolas, real-time processing, A-scan analysis**

I. Introduction

Non-invasive subsurface analysis has been widely explored in the literature [1]. Electromagnetic (EM) wave-based methods are commonly used for locating underground cables [2], but they are limited to materials that induce a magnetic field. Ground Penetrating Radar (GPR), relying on EM wave reflections, provides a general-purpose solution capable of detecting utilities, pipelines, and other subsurface structures [3]. Detecting coherent reflections in GPR data remains challenging due to surface reflections, noise, clutter, and overlapping signals. In practice, identifying hyperbolic patterns often requires manual interpretation, which is time-consuming and prone to operator variability, limiting real-time applicability.

To address these challenges, we propose a method that detects reflections directly in the A-scans as they arrive. It works in three steps:

- **A-scan region partitioning:** each trace is split into regions bounded by consecutive extrema, and each region is described by its polarity pattern, extrema spacing and central amplitude.
- **Similarity-based tracking:** a region is followed across neighbouring traces as long as its features change gradually, which traces out the reflection.
- **Parabolic approximation:** the tracked trajectory is fitted with a parabola whose apex gives the position of the buried object.



Fig. 1: Parabolic fitting of subsurface reflections on B-scan data from a real-world site.

Unlike Hough-transform and clustering methods, the method does not threshold the B-scan into a binary image and does not search over hyperbola shapes, and unlike learning-based detectors it needs no labelled training data.

Contributions: The main contributions of this work are:

- a region-based representation of A-scans that keeps the polarity and spacing of each reflection cycle,
- a tracking and parabola-fitting scheme with explicit, physically motivated constraints,
- a streaming implementation that processes an A-scan in 0.07 ms, well below the 100 ms acquisition interval, and
- an evaluation on 194 labelled field surveys against C3 and Hough-transform baselines, using a held-out test set and standard detection metrics.

The paper is organized as follows: Section II reviews related work, Section III presents the methodology, Section IV details experiments and results, and Section V concludes the paper.

II. Related Work

A. Hyperbola-Based Detection in GPR Data

Detection of subsurface objects in Ground Penetrating Radar (GPR) data is commonly based on identifying hyperbolic signatures in B-scan images, which arise from point-like or cylindrical buried targets as the antenna moves laterally across the surface [4]. Classical approaches rely on geometric curve detection, most notably Hough Transform–based methods, which search a high-dimensional parameter space to identify hyperbolas [5]. Their cost grows with the size of the parameter space, and they rely on a clean edge or binary image, which is difficult to obtain in cluttered field data [6].

B. Clustering and Hyperbola Fitting Approaches

To improve robustness, clustering-based methods were introduced to group coherent signal responses prior to curve fitting. The Column-Connection Clustering (C3) algorithm is a representative example, enabling real-time hyperbola recognition by combining region clustering with explicit hyperbola fitting [7]. C3 links thresholded pixels column by column and uses a small trained classifier to decide which clusters are hyperbolic, so it depends on both a binarisation threshold and labelled training data.

C. Parabolic Approximation of Hyperbolic Signatures

To reduce model complexity, several studies approximate hyperbolic reflections with parabolas. A point-like target at position x_0 and depth d_0 produces the two-way travel time

$$\tau(x) = \frac{2}{v}\sqrt{d_0^2 + (x - x_0)^2}, \tag{1}$$

where x is the antenna position along the survey line and v is the wave velocity in the ground. Near the apex, a second-order Taylor expansion gives

$$\tau(x) \approx \tau_0 + \frac{(x - x_0)^2}{v\, d_0}, \qquad \tau_0 = \frac{2d_0}{v}, \tag{2}$$

which is a parabola in x with its minimum at the target position [8]. Since time (or depth) increases downwards in a B-scan, this parabola appears as the familiar inverted arc. The approximation keeps the information needed for localization and has been used for apex estimation and target localization in GPR applications [8], [9].

Recent learning-based approaches employ convolutional neural networks and transformer architectures to detect hyperbolic signatures directly from B-scan images [10]–[12]. Although these methods can achieve strong detection performance, they typically require labelled training data and may incur substantial computational costs, depending on the model and deployment hardware.

In contrast, the proposed method operates directly on incoming A-scans, representing local reflection structures as regions defined by consecutive extrema. By preserving waveform polarity, extrema spacing, and amplitude characteristics, the method tracks structurally similar regions across adjacent traces without requiring explicit hyperbola detection during the tracking stage. Inverted parabolic approximation is subsequently used to estimate utility locations from the resulting trajectories. This region-based formulation avoids image-level detection and provides a traceable link between each estimated utility location and the regions contributing to its detection.

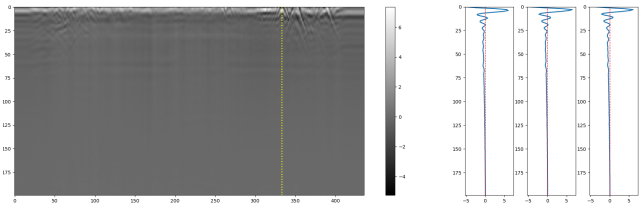


Fig. 2: GPR B-scan data showing inverted parabolic signatures generated by buried objects.

III. Methodology

A. GPR Background

To explain the proposed method for detecting inverted parabolas in GPR data, it is useful to first define *A-scan* and *B-scan*.

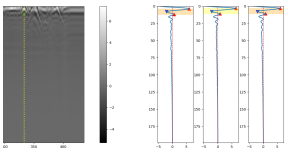
Ground Penetrating Radar (GPR) transmits electromagnetic (EM) waves into the subsurface and records the reflected signals. In a homogeneous medium, reflections are minimal, producing near-zero received amplitudes. However, buried objects or material interfaces with contrasting dielectric properties cause impedance discontinuities, reflecting EM waves and generating high-amplitude responses.

A single measurement at a fixed spatial location is called an *A-scan*, representing the signal amplitude as a function of time or depth. Visualizing these amplitudes as pixel intensities produces a one-dimensional profile. Concatenating multiple A-scans along a survey path generates a *B-scan*, a two-dimensional representation with the horizontal axis corresponding to the spatial position along the scan path and the vertical axis representing time or depth.

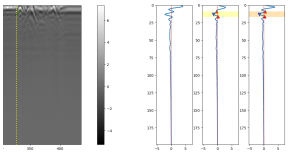
Point-like buried objects typically appear as *inverted parabolic signatures* in B-scans due to the varying propagation distance between the antenna and the object along the survey path. The apex occurs when the antenna is directly above the object, with reflections delaying symmetrically as the antenna moves away. Figure 2 shows an example B-scan and corresponding A-scan profiles at three consecutive spatial positions.

B. Method Intuition

Inverted parabolic signatures arise from consistent reflection patterns across adjacent A-scans. Strong reflections appear as prominent extrema at specific depths, and neighboring traces



(a) Single region highlighted



(b) Non-matching regions in adjacent A-scan

Fig. 3: Highlighted regions in A-scans for tracking.

exhibit similar extrema shifted gradually. Tracking these extrema across traces forms coherent trajectories corresponding to inverted parabolas in the B-scan.

The key idea is to represent local oscillatory structures in each A-scan as compact *regions*, track these regions across adjacent traces using structural similarity, and approximate the resulting trajectories with parametric parabolic models.

C. Region Definition and Representation

Let the A-scan signal at trace t be denoted as $s_t(z)$, where z is the depth index. Local extrema are extracted as

$$\mathcal{E}^t = \{e_1^t, e_2^t, \ldots, e_{N_t}^t\}, \tag{3}$$

where N_t is the number of extrema, sorted by increasing depth. To ensure full segmentation, boundary points are included:

$$\mathcal{E}^t \leftarrow \mathcal{E}^t \cup \{0, Z_{\max}\}. \tag{4}$$

A **region** is defined using three consecutive extrema:

$$R_i^t = \{e_{i-1}^t, e_i^t, e_{i+1}^t\}, \quad i = 2, \ldots, N_t - 1. \tag{5}$$

Each region corresponds to one oscillation cycle (min–max–min or max–min–max). Figure 3a highlights a sample region.

Each region R_i^t is described by the feature vector

$$\mathbf{f}_i^t = \left[\pi_i^t,\ \Delta z_1^t,\ \Delta z_2^t,\ A_i^t\right], \tag{6}$$

where π_i^t denotes the polarity pattern (max–min–max or min–max–min), $\Delta z_1^t = e_i^t - e_{i-1}^t$ and $\Delta z_2^t = e_{i+1}^t - e_i^t$ denote the spacings between consecutive extrema, and A_i^t is the amplitude of the central extremum. These features characterize the shape and strength of a local reflection cycle and provide the basis for tracking regions across adjacent traces.

D. Region Tracking Across Traces

Tracking begins from automatically selected *seed regions*. Every extremum with absolute amplitude above $A_{\min}$ within the depth interval $[z_{\min}, z_{\max}]$ seeds a region. The lower depth bound excludes the direct wave near the top of the trace. For each parent region R_i^t, a candidate in the neighbouring trace is constructed around the same-polarity extremum closest in depth to the parent's central extremum, together with the nearest surrounding extrema required to preserve the polarity pattern π.

A candidate is accepted only if it satisfies the following conditions:

$$\left|\frac{\rho^{(c)}}{\rho^{(p)}} - 1\right| \le \tau_s, \quad \frac{|A^{(c)} - A^{(p)}|}{|A^{(p)}| + \epsilon} \le \tau_a, \quad |e^{(c)} - e^{(p)}| \le \delta, \tag{7}$$

where $\rho = \Delta z_1/\Delta z_2$ is the spacing ratio, e is the depth of the central extremum, superscripts (p) and (c) denote the parent and candidate regions, respectively, and ϵ prevents division by zero. The first two conditions constrain changes in waveform shape and amplitude, while the third limits displacement between adjacent traces to reduce mismatches between distinct reflectors.

Tracking proceeds forward and backward from each seed until no candidate satisfies the acceptance criteria (Fig. 3b). Because a point-like utility produces a reflection trajectory of limited lateral extent, trajectories longer than L traces are discarded as likely layers or surface clutter.

E. B-scan Representation

Each tracked region is represented by its central extremum, forming a set of points

$$\mathcal{P} = \{(t_k, z_k)\}, \tag{8}$$

where t_k is the trace index and z_k is the depth of the central extremum. These points collectively represent the trajectory of coherent reflections across traces. In the B-scan image, the corresponding region points are highlighted as green markers, as illustrated in Fig. 3b.

F. Parabola Fitting and Target Detection

Throughout the paper, t is the trace index along the survey line and z is the depth sample, which increases downwards as in the B-scan display. In these coordinates a reflection from a point target is a parabola that opens towards larger z (Section II), i.e. $z = at^2 + bt + c$ with $a > 0$; it only looks inverted because the depth axis points down.

The depth sequence z_k of a trajectory is first smoothed with a second-order Savitzky–Golay filter. The smoothed sequence is used only to locate the apex, where it changes from decreasing to increasing, and to split long trajectories into separate rise–fall segments. The quadratic model is then fitted by least squares to the original points of the segment that contains the apex:

$$[a, b, c] = \arg\min_{a,b,c} \sum_k \left(z_k - (at_k^2 + bt_k + c)\right)^2. \tag{9}$$

The apex (t_0, z_0) is taken as the shallowest tracked point of the segment, and the detection is kept only if $a > \alpha$, which rejects flat trajectories from horizontal layers. The detected reflection is reported as

$$z = a(t - t_0)^2 + z_0, \tag{10}$$

where t_0 gives the position of the buried object along the survey line and z_0 the time of its reflection.

G. Candidate Selection and Streaming

Several seeds on the same reflection usually produce nearly the same parabola, so parabolas that are shifted copies of an accepted one, with apex and end points moved by at most 10 traces, are removed. Parabolas whose apex lies less than m samples below the direct wave are also removed, since this

band contains most of the surface clutter. The direct-wave time is estimated in each batch as the sample with the largest mean absolute amplitude among the first 15 samples.

The method processes the incoming A-scans in batches. Each A-scan is converted into regions as soon as it arrives. Every 50 traces, seeds from the latest batch, extended by 25 traces on each side, are tracked and fitted, and parabolas with their apex inside the batch are reported. Only the last 500 traces are kept in memory. All parameters are listed in Table I.

IV. Experimental Setup and Evaluation

A. GPR System and Dataset

Data were recorded with a Noggin 250 MHz GPR system (Sensors & Software Inc.), interfaced through the NIC-500N controller and read with the SPIDAR SDK. A-scans were recorded every 0.1 s, each with 200 samples at 400 ps (an 80 ns window) and 8-fold stacking. Assuming a wave velocity of about 0.1 m/ns, the window corresponds to roughly 4 m of depth. The dataset contains 194 surveys with 91,029 A-scans over an outdoor brick-paved surface.

B. Ground-Truth Annotation and Matching

Each trace is labelled as lying over a utility or not, giving 381 labelled utilities in total; 18 surveys contain none. Traces were labelled using as-built utility drawings together with visual inspection of the B-scan images. Labels mark the position of a utility along the survey line but not its depth, so detections are matched to ground truth by position only: a detected curve is a true positive if it overlaps the trace range of a labelled utility, and a curve overlapping more than one utility is assigned to the one whose trace range is closest to the curve's apex. Several detections matched to the same utility count once rather than as separate false positives. Only detections that overlap no labelled utility are treated as false positives, and among these, detections less than 15 traces apart are merged into a single false positive so that one noisy region is not counted repeatedly.

C. Baselines

We compare with two classical detectors, both re-implemented by us and run on the same batches as the proposed method. **C3** [7] removes the background, thresholds each polarity into a binary image, links pixel runs column by column into clusters, and fits a hyperbola to each cluster. The original method identifies hyperbolic clusters with a small trained classifier; since its exact features and training procedure are not available to us, our re-implementation instead accepts a cluster using explicit geometric criteria – a small fit error, a physically plausible shape, and both arms of the hyperbola present – intended to approximate the same decision, and we refer to it throughout as a modified baseline rather than an exact reproduction. The **Hough transform** detector [5], [6] votes for hyperbolas $(z - z_d)^2 = (z_0 - z_d)^2 + k^2(t - t_0)^2$ over apex position t_0, apex time z_0, and shape k, where z_d is the direct-wave time, using local extrema of the background-removed B-scan as feature points; each point only votes for curves whose slope matches the local slope of the reflection, which stops horizontal layers from producing wide false hyperbolas. For a fair comparison, both baselines use the same seed rule (a detection must pass close to a seed extremum) and the same curvature limit as the proposed method.

D. Evaluation Protocol

The surveys were split into a development set (92 surveys, 179 utilities) and a test set (101 surveys, 197 utilities, 48,122 traces), assigned alternately in date order so that both sets cover all recording days. The development set was used to choose the tuned parameters of each method, by picking the values with the highest F1. Each method was then run on the test set only once, with these values fixed. For the proposed method, tuning chose the selection rule, the lower bound of the seed depth range and the direct-wave margin (marked with * in Table I); all other parameters keep the values of the original algorithm. For the baselines, tuning chose their detection thresholds and search depth.

For each method we report precision, recall and F1. Because these scores depend on which surveys happen to be in the test set, we also give a 95% confidence interval: we draw 2000 random test sets of the same size from the 101 test surveys, allowing repeats, recompute the scores for each, and report the range covered by the middle 95% of the results. We also report the number of false positives per 100 traces and the number of utilities missed per survey.

These scores count a detection as a hit whenever its curve overlaps a utility, even if most of the curve lies elsewhere. A second score, trace-level F1, checks each trace separately, so a curve much wider than the utility it detects is penalized for the traces it wrongly covers. Since A-scans are recorded at fixed time intervals rather than fixed distances, results are given per trace and per survey rather than per meter.

E. Results

Table II gives the test results, and Fig. 4 shows a typical test survey. The three methods reach similar F1 scores and their confidence intervals overlap, so the results do not show a clear overall winner. Instead, they differ in how they trade precision against recall. The proposed method finds the most utilities (recall 0.62) but also produces the most false positives. C3 is the most precise (0.83), has the fewest false positives (0.04 per 100 traces) and the highest trace-level F1 (0.52), but misses about one utility per survey. The Hough transform lies in between on the utility-level scores, but its trace-level F1 is clearly lower (0.38), because its fitted curves are wider than the real reflections.

The proposed method can also be set to find more utilities. On the development set, starting the seeds at sample 10 and removing the direct-wave margin raises its recall from 0.60 to 0.74, while precision drops from 0.53 to 0.42. None of the baseline configurations tried during tuning reached this recall (at most 0.63 for the Hough transform and 0.45 for C3). In utility surveys, where missing a cable costs more than

TABLE I: Parameters of the proposed method. Starred entries were chosen on the development set; all others use the values of the original algorithm and were varied only to check sensitivity (Section IV).

Parameter	Role	Value
Seed amplitude $A_{\min}$	minimum extremum amplitude to start tracking	1.4
Seed depth, lower bound $z^*_{\min}$	shallowest depth where tracking may start; excludes the direct wave	13 samples (5.2 ns)
Seed depth, upper bound $z_{\max}$	deepest depth where tracking may start	40 samples (16 ns)
Spacing tolerance τ_s	allowed change in extrema spacing between adjacent traces	0.75
Amplitude tolerance τ_a	allowed change in central-extremum amplitude between adjacent traces	0.5
Max. depth change δ	largest allowed depth shift between adjacent traces; stops a trajectory jumping to another reflector	3 samples
Max. trajectory length L	longest trajectory kept; longer ones are treated as layers or clutter	50 traces
Savitzky–Golay window (order 2)	smoothing window for locating the apex / splitting the trajectory	5 (segments), 7 (apex)
Curvature threshold α	minimum fitted curvature; flatter fits are rejected	0.015 samples/trace^2
Duplicate tolerance	distance within which two fitted parabolas count as one detection	10 traces
Direct-wave margin m^*	minimum depth below the direct wave for an accepted apex; rejects near-surface clutter	8 samples (3.2 ns)
Batch length / overlap	traces processed per streaming batch, and look-ahead overlap before reporting	50 / 25 traces

TABLE II: Test-set results (101 surveys, 197 utilities). 95% confidence intervals for precision, recall, and F1 are shown in parentheses.

Method	Precision	Recall	F1	Trace F1	FP / 100 tr.	Missed / survey	Time (ms/A-scan)
Proposed	0.53 (0.46–0.61)	**0.62** (0.54–0.69)	0.57 (0.51–0.63)	0.49	0.22	**0.73**	**0.07**
C3 [7]	**0.83** (0.75–0.90)	0.48 (0.42–0.54)	**0.61** (0.55–0.66)	**0.52**	**0.04**	1.01	0.36
Hough	0.53 (0.45–0.61)	0.57 (0.50–0.64)	0.55 (0.49–0.60)	0.38	0.21	0.83	0.42

checking an extra detection, this high-recall setting may be the better choice.

Most false positives of the proposed method are short arcs just below the direct wave, caused by near-surface clutter, such as those on the right of Fig. 4(a). Utilities are missed mainly when their reflection is too weak for any extremum to exceed the seed amplitude; U1 in Fig. 4 is missed by all three methods.

F. Processing Time and Memory

All timings were measured on a desktop PC (Intel Core i7-13700, 16 GB RAM), running each method single-threaded in Python. The proposed method needs 0.07 ms per A-scan on average, compared with 0.36 ms for C3 and 0.42 ms for the Hough transform. A new A-scan arrives every 100 ms, so all three methods are far faster than needed. The slowest batch of the proposed method took less than 15 ms, against 5 s available, and it used less than 4 MB of memory, compared with less than 1 MB for C3 and up to 68 MB for the Hough transform.

Processing time is different from detection delay, the time between the antenna passing a target and the detection being shown. Traces are processed in batches of 50, and a batch is only finished once 25 more traces have arrived, so that a curve starting near the end of the batch can be tracked to its end. A target therefore waits for the rest of its batch plus these 25 traces: about 2.6 s if it lies near the end of the batch and up to 7.5 s if it lies near the start, about 5 s on average. This delay comes from the batch size, not from the computation, and is the same for all three methods.

G. Parameter Sensitivity

To check how much the results depend on the parameter values, each parameter of the proposed method was changed on its own, on the development set, while the others were kept at their values in Table I. Changing τ_s (0.5–1.0), τ_a (0.3–0.7), α (0.005–0.03), δ (2–4 samples), L (40–60 traces) or m (6–10 samples) kept F1 between 0.55 and 0.59, so the method is not sensitive to these choices. The seed amplitude matters most, because it sets the balance between precision and recall: lowering it from 1.4 to 1.2 raised recall from 0.60 to 0.70 but lowered precision from 0.53 to 0.42, while raising it to 1.6 raised precision to 0.71 and lowered recall to 0.53.

H. Limitations

A-scans are recorded at fixed time intervals, so the horizontal axis of the B-scan counts traces rather than metres. If the operator walks faster or slower, reflections become narrower or wider along this axis. Because δ, L and α are measured in traces, their best values depend on walking speed; recording by distance, for example with a wheel encoder, would remove this dependence. The labels mark only where a utility lies along the survey line, so the accuracy of the estimated depth could not be checked. Finally, both baselines are our own re-implementations, and our C3 uses geometric checks instead of its original trained classifier.

V. Conclusion

We presented a method that detects buried utilities in GPR data directly from the incoming A-scans. Each A-scan is split into regions bounded by consecutive extrema, these regions are tracked across neighbouring traces using their polarity, spacing and amplitude, and the resulting trajectories are fitted with parabolas to locate the utilities. The method needs no training data and takes 0.07 ms per A-scan on average, far less than the 100 ms between A-scans.

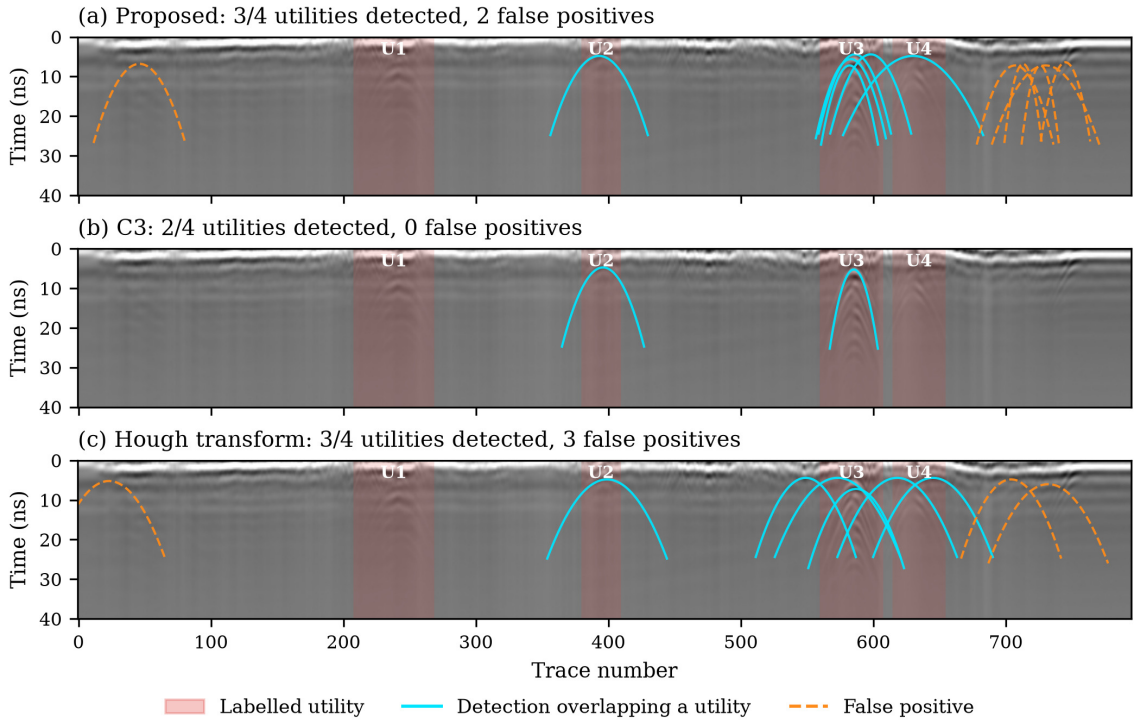


Fig. 4: Detections on a test survey (795 traces, 4 labelled utilities) by (a) the proposed method, (b) C3 and (c) the Hough transform, each with its parameters tuned on the development set. Shaded bands mark labelled utilities; solid curves overlap a utility, dashed curves are false positives.

On 101 held-out surveys with 197 labelled utilities, the method reached an F1 score of 0.57, comparable to C3 (0.61) and the Hough transform (0.55). It found the most utilities (recall 0.62) and missed fewer per survey than either baseline (0.73, compared with 1.01 for C3 and 0.83 for the Hough transform), at a lower computational cost. Its main weakness is precision: at 0.53 it is well below C3 (0.83), mainly because of false positives from near-surface clutter. Since a missed utility is far more costly in the field than a false alarm, this balance can be preferable for utility detection.

Future work will focus on rejecting near-surface clutter, recording A-scans by distance rather than time to remove the dependence on walking speed, and reporting detections as soon as a trajectory ends instead of waiting for the batch to finish.

AI Usage Declaration

The authors acknowledge the use of AI-assisted tools during the preparation of this manuscript. These tools were employed to improve language clarity, grammar, and overall readability. All technical content, analysis, and results were independently developed and verified by the authors.

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